

MACHINE LEARNING - Set 2: Scenario Based Questions

Topic: Identify Problem Type & Outline Logic

1. A bank wants to predict whether a loan applicant will default based on credit score, income, and past loan history.

Type: Supervised – Classification (Binary)

Logic:

1. Collect data: credit score, income, loan history, default label (0/1)
2. Preprocessing: Handle null values, One-Hot Encode categorical, StandardScaler for income
3. Split: Train/Test 80/20
4. Model: Logistic Regression / Random Forest Classifier
5. Evaluate: Accuracy, Precision, Recall, F1-Score, ROC-AUC

2. A retail store wants to predict the demand for different products to optimize inventory levels.

Type: Supervised – Regression

Logic:

1. Collect data: product id, past sales, price, season, offer, demand quantity
2. Preprocessing: Handle seasonality, Encode product category, Normalize
3. Split: Train/Test by time (TimeSeriesSplit)
4. Model: XGBoost Regressor / Linear Regression / ARIMA
5. Evaluate: RMSE, MAE, R2-Score

3. A factory wants to detect whether a manufactured product is defective based on sensor readings and quality control data.

Type: Supervised – Classification

Logic:

1. Collect data: temperature, vibration, pressure, QC label (defective/good)
2. Preprocessing: Remove sensor noise, Handle missing, Normalize
3. Split: Train/Test 80/20
4. Model: SVM / Random Forest Classifier
5. Evaluate: Confusion Matrix, Precision, Recall

4. A healthcare provider wants to analyze patient symptoms and classify them into different disease categories.

Type: Supervised – Classification (Multi-class)

Logic:

1. Collect data: symptoms, age, test reports, disease label
2. Preprocessing: Encode symptoms, Handle nulls, Scale
3. Split: Train/Test 80/20 with stratification
4. Model: Decision Tree / Naive Bayes / Random Forest
5. Evaluate: Accuracy, F1-Score, Classification Report

5. An e-commerce company wants to identify and remove fake reviews posted by bots or fraudsters.

Type: Supervised – Classification (NLP)

Logic:

1. Collect data: review text, rating, user IP, fake/real label
2. Preprocessing: Text cleaning, TF-IDF / BERT Embedding, Remove stopwords
3. Split: Train/Test 80/20
4. Model: Logistic Regression / LSTM / BERT Classifier
5. Evaluate: Precision, Recall, F1-Score

6. A financial firm wants to predict stock price movements based on historical price data and market indicators.

Type: Supervised – Regression / Time Series Forecasting

Logic:

1. Collect data: open, close, volume, RSI, moving average
2. Preprocessing: Create lag features, Normalize, Handle missing
3. Split: Train/Test by time, don't shuffle
4. Model: LSTM / XGBoost Regressor / ARIMA
5. Evaluate: RMSE, MAPE, MAE

7. A social media platform wants to detect fake user accounts based on user activity and profile data.

Type: Supervised – Classification

Logic:

1. Collect data: no of posts, followers, account age, fake/real label
2. Preprocessing: Feature engineering, Scale numeric, Encode
3. Split: Train/Test 80/20
4. Model: Random Forest / XGBoost Classifier

5. Evaluate: Accuracy, ROC-AUC, F1-Score

8. A marketing agency wants to segment customers into different groups based on their purchasing behavior.

Type: Unsupervised – Clustering

Logic:

1. Collect data: purchase frequency, amount, recency, product type
2. Preprocessing: RFM Analysis, StandardScaler
3. Model: K-Means Clustering
4. Evaluate: Elbow Method to find K, Silhouette Score
5. Outcome: Create 3-4 customer groups for targeting

9. A geospatial research team wants to analyze satellite images to classify different land types (forest, water, urban).

Type: Supervised – Classification (Image Classification)

Logic:

1. Collect data: satellite images with labels (forest, water, urban)
2. Preprocessing: Resize to 224x224, Normalize, Data Augmentation
3. Split: Train/Test/Validation 70/15/15
4. Model: CNN / ResNet / EfficientNet
5. Evaluate: Accuracy, Confusion Matrix, IoU

10. A streaming service wants to predict which users are likely to cancel their subscriptions.

Type: Supervised – Classification (Churn Prediction)

Logic:

1. Collect data: watch time, last login, plan type, churn label (yes/no)
2. Preprocessing: Handle imbalance (SMOTE), Encode plan, Scale
3. Split: Train/Test 80/20
4. Model: Logistic Regression / XGBoost Classifier
5. Evaluate: Precision, Recall, F1-Score, Churn Rate